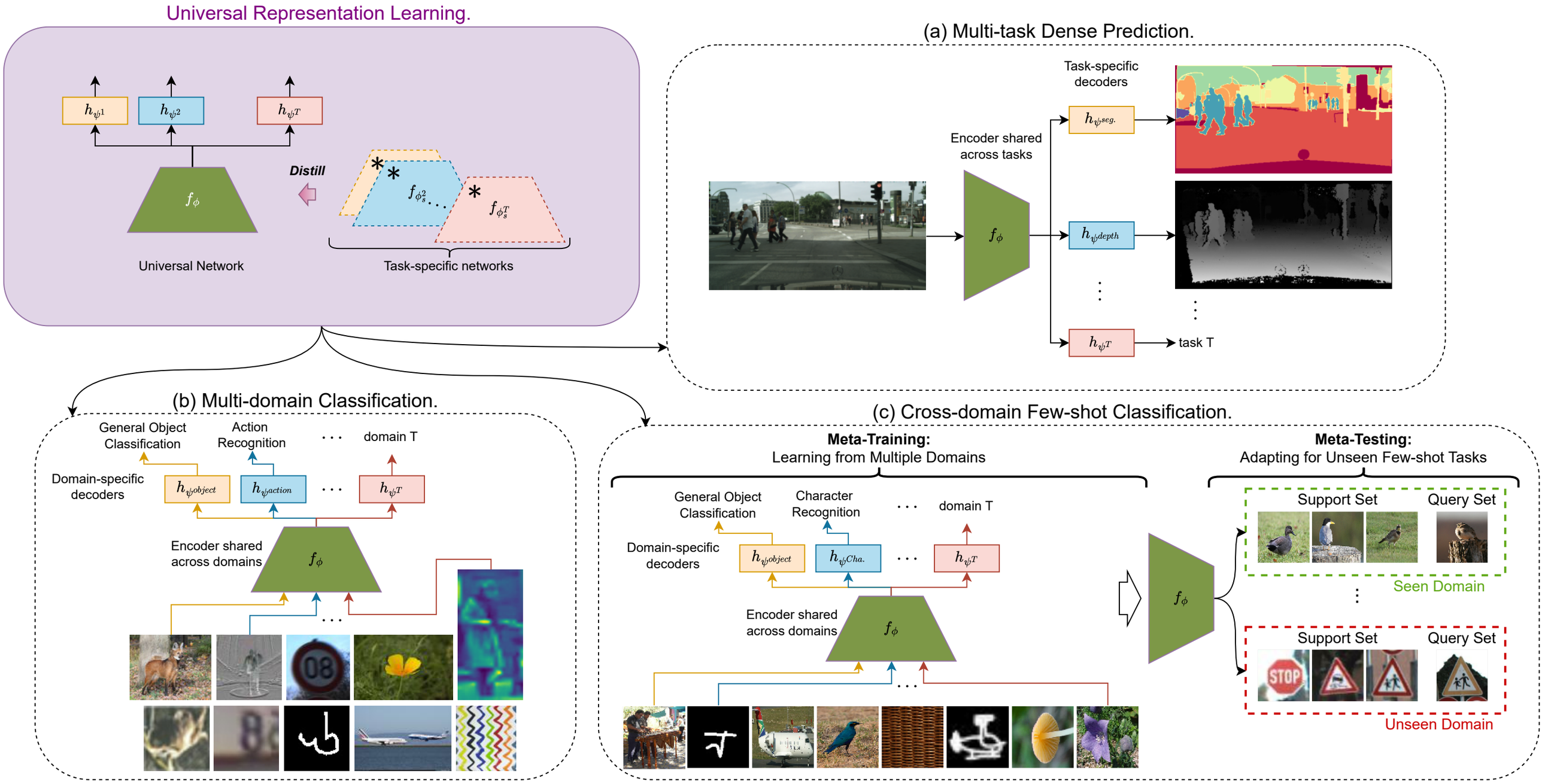
Universal Representation Learning.
h_{ψ^1}
h_{ψ^2}
h_{ψ^T}
f_ϕ
Distill
$f_{\phi_s^2}$
$f_{\phi_s^T}$
Universal Network
Task-specific networks
(a) Multi-task Dense Prediction.
Task-specific decoders
Encoder shared across tasks
$h_{\psi^{seg.}}$
$h_{\psi^{depth}}$
h_{ψ^T}
task T
(b) Multi-domain Classification.
General Object Classification
Action Recognition
domain T
Domain-specific decoders
$h_{\psi^{object}}$
$h_{\psi^{action}}$
Encoder shared across domains
(c) Cross-domain Few-shot Classification.
Meta-Training:
Learning from Multiple Domains
General Object Classification
Character Recognition
domain T
Domain-specific decoders
$h_{\psi^{object}}$
$h_{\psi^{Cha.}}$
Encoder shared across domains
Meta-Testing:
Adapting for Unseen Few-shot Tasks
Support Set
Query Set
Seen Domain
Support Set
Query Set
Unseen Domain